8 PHOTOGRAMMETRY

GENERAL DEFINITIONS

1. Which of the following best defines the term *fiducial marker* as it is used in photogrammetry?

(A) A fiducial marker is one of the (usually) two images on a photograph that define the axis of the tilt.

(B) A fiducial marker is a premarked target set on the ground to balance the stereo model.

(C) A fiducial marker is one of the (usually) four objects connected to the camera's interior that form images on the negative as each photograph is taken.

(D) A fiducial marker is the plane that defines the camera's focal length.

Solution: The images of four fiducial markers fixed inside the camera appear on each photograph. These markers allow consistent orientation of the photographs.

Answer is (C)

2. Which of the following statements best describes the Porro-Koppe principle?

(A) Distortion caused by imperfections in the lens system of a plotter can be minimized if the imperfections have the same distortion as the lenses of the camera used to create the photographs.

(B) The image plane, object plane, and the plane of the lens of a direct-focusing projector intersect in the same line to ensure sharp focus.

(C) Irregular changes in brightness are proportional to the refractive index between the object and the image.

(D) The focal length of the camera, the flight height of the aircraft, and the elevation of the point determine the scale.

Solution: According to the law of reversibility of light paths, the effect of lens distortion in the camera is reversed in the plotting instruments if identical imperfections exist in both lens systems.

Answer is (A)

3. Which of the following best describes the term *principal point* as it is used in photogrammetry?

(A) It is a point on the ground used as the basis for the determination of scale and orientation.

(B) It is synonymous with focal point.

(C) It is a point that appears on both photographs of a stereo pair.

(D) It is the point at which the lines through opposing sets of fiducial marks intersect.

Solution: The principal point is the intersection of the optical axis of the camera and the principal surface and is indicated by the intersection of the lines through the fiducial markers.

Answer is (D)

4. Which of the following best describes an orthophotograph?

(A) a stereo pair of aerial photographs viewed together to represent the three-dimensional nature of the area photographed

(B) a mosaic of aerial photographs at the scale of a standard quad, 1:24,000

(C) a perspective photograph from which the displacements caused by tilt and relief have been removed

(D) an aerial photograph in which the horizon is visible

Solution: A stereoscopic plotting instrument theoretically presents a model of the terrain photographed when a stereo pair is properly oriented. Optical and electronic manipulation can eliminate the displacement of the points in the photographs due to tilt and relief. The result is known as an orthophotograph.

Answer is (C)

5. Which of the following best describes a vertical photograph?

(A) an aerial photograph taken with a camera whose optical axis is virtually vertical

(B) an aerial photograph taken with a camera whose optical axis is virtually horizontal

(C) an aerial photograph in which the horizon is visible

(D) an aerial photograph taken with a camera whose optical axis has been tilted but in which the horizon is not visible

Solution: A high oblique aerial photograph is one in which the horizon is visible. A low oblique photograph is taken with a tilted camera, but the horizon is not visible. A vertical photograph is made with a camera whose optical axis is nearly vertical.

Answer is (A)

6. What is the meaning of the term *air base* as it is used in photogrammetry?

(A) the mounting of the aerial camera in the aircraft

(B) the line between two air stations where overlapping photographs were taken

(C) the absolute stereoscopic parallax of a point appearing on a stereo pair of photographs as determined by the algebraic difference of the distance of the points from their respective principal points

(D) the rate of change of parallax with respect to the changes in flight height

Solution: The air base is also known as the *baseline* or the *model base*. It is the distance the plane flew between overlapping exposures. As reconstructed in the plotting instrument, it is the distance, at scale, between adjacent perspective centers.

Answer is (B)

7. Which of the following best describes the K-factor?

(A) the smallest contour interval that can be plotted at the required mapping accuracy

(B) the degree to which an image is displaced radially toward or away from the nadir point of the photograph

(C) the apparent inward or outward displacement of an image from the principal point due to lens design

(D) the ratio of the air base length to the altitude at which a pair of aerial photographs were taken

Solution: The base-height ratio is also known as the *K-factor*.

Answer is (D)

8. What is meant when an aerial photograph is said to have been rectified?

(A) the effects of tilt have been removed
(B) the photograph has been evaluated for vertical map accuracy in a stereoplotte
(C) the photograph has been transferred to a transparent positive on a glass plate
(D) the photograph has been reduced to an orthophotograph

Solution: An aerial photograph can be translated, rotated, and scaled so as to reverse the effects of the original picture-taking process, eliminating the effects of tilt, but not of relief displacement.

Answer is (A)

9. In a true vertical photograph made by a perfectly adjusted camera, which two points would be identical?

(A) the principal point and the nadir point
(B) the detail point and the pass point
(C) the tie point and the base point
(D) the picture point and the photograph center

Solution: In a photograph taken with the optical axis of the camera aligned with gravity, the point at which the line through the perspective center of the lens system intersects the photograph would coincide with the principal point.

Answer is (A)

10. What is a stereoscope?

(A) a portable optical device used to view two photographs simultaneously, creating the impression of a three-dimensional image

(B) a device used to measure the separation between two index marks on a stereo pair of aerial photographs

(C) an optical device used for measuring coordinates, both rectangular and polar, on a photographic plate

(D) a composite from a pair of slotted templates representing a stereoscopic model

Solution: A stereoscope is a small optical device that allows a viewer, with some practice, to view two-dimensional photographs simultaneously and form a mental image of the three-dimensional surface they represent.

Answer is (A)

11. Which of the following best defines tilt in photogrammetry?

(A) Tilt is the angle between the nadir point and the principal point of an aerial photograph.

(B) Tilt is the apparent deflection of the image of a straight line on the ground due to relief displacement.

(C) Tilt is the dihedral angle between a truly horizontal plane and the plane of the aerial photograph.

(D) Tilt refers to the angle used in the radial line method of relating the isocenter to other points on the photograph.

Solution: The tilt of an aerial photograph is caused by the impossibility of maintaining a truly vertical camera axis when making the exposures. The inclination of the optical axis of the camera from the vertical is usually expressed as the angle between the plane of the photograph and the horizontal plane. The effects of tilt can be compensated for in stereoscopic plotting instruments by the rotation and movement of projectors to reflect the position of the aerial camera when the photographs were made.

Answer is (C)

12. What is the difference between x-tilt and y-tilt?

(A) x-tilt is the angle of rotation around the axis perpendicular to the flight line, while y-tilt is the angle of rotation around the axis of the flight line.

(B) x-tilt is the angle of rotation around the axis of the flight line, while y-tilt is the angle of rotation around the axis perpendicular to the flight line.

(C) x-tilt is the angle of rotation around the axis described by the principal point and the perspective center, while y-tilt is the angle of rotation around the axis perpendicular to the flight line.

(D) x-tilt is the angle of rotation around the axis perpendicular to the flight line, while y-tilt is the angle of rotation around the axis described by the principal point and the perspective center.

Solution: The x-axis is assumed to be the line described by fiducial marks that most closely matches the flight line. The y-axis is nearly perpendicular to the flight line. In other words, when the nose of the aircraft moves down, the result would be described as a positive y-tilt.

Answer is (B)

13. Which of the following assumptions is the basis for the radial line method of plotting on aerial photographs?

(A) The photographs have such insignificant angles of tilt that they are assumed to be truly vertical photographs.

(B) The principal point, the plumb point, and the pass point are identical.

(C) The relief displacement in the photograph is equal at each edge.

(D) The x-parallax in the photograph is zero.

Solution: In the radial line method of plotting, the angles drawn from the radial center to other points on an aerial photograph are assumed to equal the corresponding angles on the ground. The method is based on the assumption that the camera was perpendicular to the horizontal plane at the moment of exposure. This condition is almost never met, but the magnitude of the tilt is often not significant enough to negate the plotting method.

Answer is (A)

14. What is the isocenter of an aerial photograph?

(A) the point on the ground directly beneath the optical center of the lens at the moment of the exposure

(B) the point at the intersection of three planes: the truly horizontal plane, the plane of the tilted photograph, and the vertical plane that contains the optical axis of the camera

(C) a point whose horizontal and vertical positions are known from photogrammetric methods that is used for the absolute orientation of other photographs

(D) the foot of the perpendicular from the perspective center to the plane of the photograph

Solution: The isocenter is approximately midway between the nadir point and the principal point on a tilted aerial photograph. Its distance from the principal point is equal to the focal length multiplied by the tangent of half the angle of the tilt.

Answer is (B)

15. What sort of composite of aerial photographs involves trimming the edges of the individual photographs?

(A) a controlled mosaic

(B) a planimetric map

(C) an uncontrolled mosaic

(D) a topographic map

Solution: The construction of a controlled mosaic or photo map requires cutting the edges of the prints along features that would reproduce as lines anyway.

Answer is (A)

FORMATS AND CAMERAS

16. What are the typical dimensions of the portion of the negative within the focal plane of an aerial camera used for mapping?

(A) 8 in × 10 in
(B) 12 in × 12 in
(C) 4 in × 5 in
(D) 9 in × 9 in

Solution: The usual format for aerial photographs used in topographic mapping is 9 in × 9 in (230 mm × 230 mm).

Answer is (D)

17. What is the term used to describe a positive aerial photograph printed on a transparent plate?

(A) a dioptric
(B) a trimetrogon
(C) a strip photograph
(D) a diapositive

Solution: Aerial photographs are printed on glass plates in preparation for their projection in a stereoscopic plotting instrument or comparator. These diapositives are sometimes produced at a reduced scale, using lenses that compensate for the distortion of the camera lens used to make the photographs.

Answer is (D)

18. Which of the following is not a typical component of an aerial camera?

(A) a magazine
(B) a cone
(C) a diaphragm
(D) a collimator

Solution: A collimator is used to calibrate the camera, but is not part of the camera itself.

Answer is (D)

19. In a photogrammetric camera lens, which focal length specification would most emphasize the effects of parallax?

(A) an $8\frac{1}{4}$-in focal length
(B) a 12-in focal length
(C) a 6-in focal length
(D) a 10-in focal length

Solution: The magnification of parallax is a desirable characteristic in aerial photographs used for topographic purposes. The measurement of parallax is a critical factor in plotting topographic maps. A relatively short focal length and a wide-angle field is preferred for such work. The 6-in focal length wide-angle lens is typical.

Answer is (C)

20. What sort of shutter is used on most aerial cameras?

(A) a capping shutter
(B) a focal-plane shutter
(C) a louvre shutter
(D) a between-the-lens shutter

Solution: While the focal-plane shutter is capable of faster speeds than the between-the-lens shutter, this efficiency is offset by the distortion introduced when such a shutter is used in a moving aircraft. The between-the-lens shutter exposes the whole negative at once and is preferred over the focal-plane shutter, which exposes the negative from one end to the other by moving across it.

Answer is (D)

21. What is the function of an intervalometer?

(A) It controls the loading of the film magazine.
(B) It controls the relationship between the object plane, lens, and image plane of the camera.
(C) It controls the splitting of the light passing through the lens of the camera into two beams.
(D) It controls the shutter of the camera.

Solution: Once it is set, the intervalometer operates the shutter of the camera automatically, making exposures that take the speed of the airplane into account to assure photographs with a predetermined overlap.

Answer is (D)

22. Which component unique to a reseau camera is used to minimize the distortion in its aerial photographs?

(A) a super-aviogon lens
(B) a vacuum pump film flattening system
(C) a goniometer
(D) a rectangular grid etched on glass

Solution: In a reseau camera, a rectangular grid is superimposed on the image produced by the exposure. The distortion of the photograph can then be minimized by restoring the grid to its proper shape and size.

Answer is (D)

23. For a given format size, as the focal length of a camera decreases, which of the following occurs?

(A) Radial distortion increases.
(B) Tangential distortion decreases.
(C) Spherical aberration increases.
(D) The angular field of view increases.

Solution: The diagonal or angular field of view for a given format increases as the focal length of the camera lens decreases. This is significant in photogrammetry since the shorter the focal length of the lens used, the wider the ground coverage that can be achieved on a single photograph.

Answer is (D)

24. Why will an aerial camera that has not been corrected for the crab angle cause a reduction in the ground coverage of the resulting photographs?

(A) The sides of the photographs will not be parallel to the flight line.
(B) The endlap of the photographs will exceed 60%.
(C) The photographs will have an excessive x-tilt.
(D) The photographs will have an excessive y-tilt.

Solution: Side winds can cause the heading of an aircraft to deviate from its actual direction of travel. The angle between the longitudinal axis of the aircraft and its flight line is known as the *yaw* or *crab angle*. When an aerial camera is not corrected for this effect, the sides of the resulting photographs are not parallel with the direction of flight, and the ground coverage can be significantly reduced.

Answer is (A)

25. Which of the following types of aerial camera is most often used for topographic mapping purposes?

(A) continuous-strip cameras
(B) multilens convergent cameras
(C) multicamera systems
(D) single-lens cameras

Solution: A single-lens precision camera is preferred for topographic mapping. It provides a combination of short focal length and wide angle for high geometric picture quality.

Answer is (D)

26. What is the angular field of view for a single-lens aerial camera with a 6-in fixed focal length and a 9-in × 9-in format?

(A) 75°
(B) 93°
(C) 122°
(D) 56°

Solution: Given the format size and focal length of a camera, the formula for calculating the diagonal field of view is

$$\alpha = 2\tan^{-1}\left(\frac{d}{2f}\right)$$

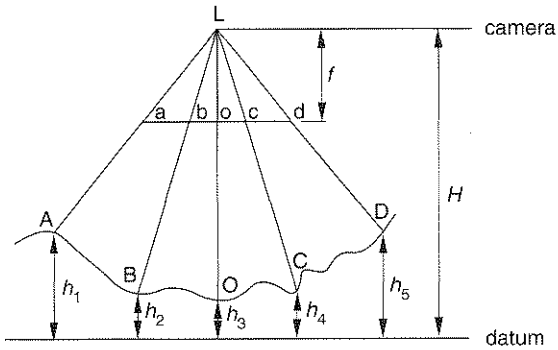
d is the length of the diagonal across the format, and f is the focal length.

$$\begin{aligned}\alpha &= 2\tan^{-1}\left[\frac{\sqrt{(9\text{ in})^2 + (9\text{ in})^2}}{(2)(6\text{ in})}\right]\\ &= 2\tan^{-1}\left(\frac{12.728}{12}\right)\\ &= 2\tan^{-1}(1.0607)\\ &= 93^\circ\end{aligned}$$

Answer is (B)

SCALE

The following two problems refer to the illustration shown.



27. A camera with a focal length of 6 in was used to make a vertical photograph at an altitude of 10,000 ft above sea level. The height of point B and point C is approximately 4500 ft above sea level in both cases. What is the approximate scale along line BC in the photograph?

(A) 1:20,000
(B) 1:11,000
(C) 1:1000
(D) 1:750

Solution: The formula for the scale at a particular elevation is

$$S_e = \frac{f}{H - h}$$

S_e is the scale at a given elevation, f is the focal length, H is the altitude of the aircraft above a datum, and h is the height of the ground station above the same datum.

$$\begin{aligned} S_e &= \frac{6 \text{ in}}{10{,}000 \text{ ft} - 4500 \text{ ft}} \\ &= \frac{0.5 \text{ ft}}{5500 \text{ ft}} \\ &= \frac{1}{11{,}000} \end{aligned}$$

The solution relies on solving the similar triangles Lbc and LBC and on presuming that line BC lies across essentially level terrain. The answer means that 1 unit on the photograph is equal to approximately 11,000 of the same units on the ground along line BC.

Answer is (B)

28. A camera with a focal length of 6 in was used to make a vertical photograph at an altitude of 3000 ft above sea level. Point B is 150 ft above sea level, and point A is 800 ft above sea level. What is the scale at points A and B in the photograph?

(A) A: 1 in = 475 ft
B: 1 in = 367 ft

(B) A: 1 in = 442 ft
B: 1 in = 363 ft

(C) A: 1 in = 546 ft
B: 1 in = 824 ft

(D) A: 1 in = 367 ft
B: 1 in = 475 ft

Solution: The formula for the scale at a particular elevation is

$$S_e = \frac{f}{H - h}$$

S_e is the scale at a given elevation, f is the focal length, H is the altitude of the aircraft above a datum, and h is the height of the ground station above the same datum. For point B,

$$\begin{aligned} S_e &= \frac{6 \text{ in}}{3000 \text{ ft} - 150 \text{ ft}} \\ &= \frac{6 \text{ in}}{2850 \text{ ft}} \\ &= \frac{1 \text{ in}}{475 \text{ ft}} \end{aligned}$$

Repeat the same calculation for point A.

$$\begin{aligned} S_e &= \frac{6 \text{ in}}{3000 \text{ ft} - 800 \text{ ft}} \\ &= \frac{6 \text{ in}}{2200 \text{ ft}} \\ &= \frac{1 \text{ in}}{367 \text{ ft}} \end{aligned}$$

Answer is (D)

29. What flight height would be required to produce vertical photographs with a scale of 1:12,000 over level terrain if the average elevation is 3300 ft and the focal length of the camera is 8 in?

(A) 10,000 ft
(B) 15,300 ft
(C) 11,300 ft
(D) 8200 ft

Solution: Again,

$$S_{\text{ave}} = \frac{f}{H - h_{\text{ave}}}$$

S_{ave} is the average scale stipulated, f is the focal length, H is the altitude of the aircraft above a datum, and h_{ave} is the average elevation of the terrain above the same datum.

$$\begin{aligned} \frac{1}{12{,}000} &= \frac{0.67 \text{ ft}}{H - 3300 \text{ ft}} \\ H - 3300 \text{ ft} &= (0.67 \text{ ft})(12{,}000) \\ H &= 8000 \text{ ft} + 3300 \text{ ft} = 11{,}300 \text{ ft} \end{aligned}$$

Answer is (C)

30. The length of a one-story building is measured as 3.52 in on a vertical photograph. A map, which has a scale of 1:24,000, shows the same building as having a length of 2.67 in. What is the engineer's scale of the photograph?

(A) 1 in = 2400 ft
(B) 1 in = 1136 ft
(C) 1 in = 4321 ft
(D) 1 in = 1517 ft

Solution: The scale of the photograph can be defined as

$$\begin{aligned}\text{scale of the photograph} &= \left(\frac{\text{distance on the photo}}{\text{distance on the map}}\right)(\text{map scale})\\ &= \left(\frac{3.52\text{ in}}{2.67\text{ in}}\right)\left(\frac{1}{24{,}000}\right)\\ &= (1.3183)\left(\frac{1}{24{,}000}\right)\\ &= \frac{1}{18{,}205}\end{aligned}$$

This scale may be read as

$$\begin{aligned}1\text{ in} &= 18{,}205\text{ in}\\ &= (18{,}205\text{ in})\left(\frac{1\text{ ft}}{12\text{ in}}\right)\\ &= 1517\text{ ft}\end{aligned}$$

Answer is (D)

31. A distance between two road intersections on a vertical photograph is 5.12 in. The same distance on a map with a scale of 1:24,000 measures 3.00 in. The elevation of the essentially level terrain in the area is 1200 ft above sea level. If the focal length of the lens in the camera used to make the photograph was 6 in, what was the altitude of the aircraft above sea level when the photograph was taken?

(A) 8230 ft
(B) 7197 ft
(C) 14,060 ft
(D) 12,000 ft

Solution: The scale of the photograph can be expressed as

$$\begin{aligned}\text{scale of the photograph} &= \left(\frac{\text{distance on the photo}}{\text{distance on the map}}\right)(\text{map scale})\\ &= \left(\frac{5.12\text{ in}}{3.00\text{ in}}\right)\left(\frac{1}{24{,}000}\right)\\ &= (1.707)\left(\frac{1}{24{,}000}\right)\\ &= \frac{1}{14{,}060}\end{aligned}$$

To find the altitude of the aircraft, solve the equation

$$S_{\text{ave}} = \frac{f}{H - h_{\text{ave}}}$$

S_{ave} is the average scale, f is the focal length, H is the altitude of the aircraft above a datum, and h_{ave} is the average elevation of the terrain above the same datum.

$$\begin{aligned}\frac{1}{14{,}060} &= \frac{0.5\text{ ft}}{H - 1200\text{ ft}}\\ H - 1200\text{ ft} &= (0.5\text{ ft})(14{,}060)\\ H &= 7030\text{ ft} + 1200\text{ ft}\\ &= 8230\text{ ft}\end{aligned}$$

Answer is (A)

32. The scale of two diapositives used to compile a manuscript in a Kelsh plotter is 1 in = 500 ft. If the focal length of the camera is 6 in and the optimum projection distance of the plotter is 760 mm, what is the optimum model scale?

(A) 1 in = 1000 ft
(B) 1 in = 500 ft
(C) 1 in = 200 ft
(D) 1 in = 100 ft

Solution: The optimum projection distance, which is the plane of best focus, is used to fix the model scale for a particular stereo plotter. The formula used is

$$\begin{aligned}\text{model scale} &= \left(\frac{\text{projection distance}}{\text{camera focal length}}\right)(\text{photo-scale})\\ &= \left(\frac{760\text{ mm}}{6\text{ in}}\right)\left(\frac{1\text{ in}}{500\text{ ft}}\right)\left(\frac{1\text{ in}}{25.4\text{ mm}}\right)\\ &= \frac{1\text{ in}}{100\text{ ft}}\end{aligned}$$

Answer is (D)

33. A multiplex plotting instrument with a fixed plotting scale has an optimum projection distance of 360 mm. If plotting is anticipated on a multiplex instrument at a scale of 1:10,000 and the average elevation of the terrain is 3000 ft, what flight height above sea level should be used?

(A) 10,000 ft
(B) 16,310 ft
(C) 12,870 ft
(D) 14,810 ft

Solution: The plotting scale is fixed in the multiplex instrument and therefore is the same as the model scale.

$$\text{optimum model scale} = \frac{\text{optimum projection distance}}{\text{flight height} - \text{average terrain elevation}}$$

The model scale is known in this case; it is 1:10,000. The optimum projection distance is more conveniently expressed in feet.

$$(360\text{ mm})\left(\frac{1\text{ in}}{25.4\text{ mm}}\right)\left(\frac{1\text{ ft}}{12\text{ in}}\right) = 1.181\text{ ft}$$

$$\frac{1}{10{,}000} = \frac{1.181\text{ ft}}{\text{flight height} - 3000\text{ ft}}$$

$$\begin{aligned}\text{flight height} - 3000\text{ ft} &= (10{,}000)(1.181\text{ ft})\\ &= 11{,}810\text{ ft}\\ \text{flight height} &= 11{,}810\text{ ft} + 3000\text{ ft}\\ &= 14{,}810\text{ ft}\end{aligned}$$

Answer is (D)

RELIEF DISPLACEMENT

34. Which of the following best defines the term *relief displacement* as it is used in photogrammetry?

(A) the radial movement of the images on a tilted photograph with respect to the isocenter when they are on the high or low side of the isometric parallel

(B) the range of points that can be focused sharply in an aerial photograph

(C) the angle at a point on the ground between the direction of gravity and the line through the point normal to a reference surface

(D) the radial displacement of a point on a photograph toward or away from the nadir caused by the elevation of the object above or below a given datum

Solution: Points above or below each other on the ground are displaced from each other on an aerial photograph. This displacement is along a line from the image points to the principal point or nadir. On the photograph, points below the datum are displaced toward the nadir.

Answer is (D)

35. Which variables determine the extent of relief displacement on an aerial photograph?

(A) the size of the format, the flight height above the datum, and the radial distance of the image point from the principal point on the photograph
(B) the overlap and the flight height above the datum
(C) the elevation of the point from the datum, the flight height over the datum, and the radial distance of the image point from the principal point on the photograph
(D) the elevation of the point from the datum, the size of the format, the flight height above the datum, and the radial distance from the principal point on the photograph

Solution: The extent of relief displacement can be defined as

$$d = \frac{rh}{H}$$

d is the relief displacement, r is the radial distance of the ground point image from the principal point, h is the elevation of the point from the datum, and H is the flight height above the datum.

Answer is (C)

36. A vertical photograph made at a flight height of 2000 ft above sea level shows a radio tower with a base elevation 540 ft above the same datum. The image of the tower has a relief displacement of 1.33 in. The distance from the photograph's principal point to the top of the tower is 5.97 in. What is the height of the tower to the nearest foot?

(A) 433 ft
(B) 240 ft
(C) 512 ft
(D) 325 ft

Solution: The most convenient datum for the calculation is the bottom of the tower at 540 ft. The flight height can then be expressed relative to that base.

$$\begin{aligned}\text{flight height} &= 2000 \text{ ft} - 540 \text{ ft} \\ &= 1460 \text{ ft (relative to the base of the tower)}\end{aligned}$$

d is the relief displacement, r is the radial distance of the ground point image from the principal point, h is the elevation of the point from the datum in feet, and H is the flight height above the datum in feet. The formula for relief displacement ($d = rh/H$) may be rearranged

$$\begin{aligned}h &= \frac{Hd}{r} \\ &= \frac{(1.33 \text{ in})(1460 \text{ ft})}{5.97 \text{ in}} \\ &= \frac{1941.80 \text{ ft}}{5.97 \text{ in}} \\ &= 325.25 \text{ ft}\end{aligned}$$

Answer is (D)

37. At the bottom of a valley, the scale of a particular vertical photograph is 1:8000. The focal length of the lens used to make the photograph is 6 in. A road intersection on the same photograph is 495 ft above the valley floor and 3.99 in from the principal point. What is the relief displacement of the road intersection with respect to the bottom of the valley?

(A) 1.52 in
(B) 0.75 in
(C) 3.21 in
(D) 0.50 in

Solution: Using the valley floor as the datum ($h = 0$), the formula may be written as

$$S = \frac{f}{H}$$

S is the scale at the valley floor, f is the focal length of the lens, and H is the altitude of the aircraft.

$$\begin{aligned}\frac{1}{8000} &= \frac{0.5 \text{ ft}}{H} \\ H &= (0.5 \text{ ft})(8000) \\ &= 4000 \text{ ft}\end{aligned}$$

The altitude of the aircraft when the photograph was taken was 4000 ft above the valley floor.

Next, the relief displacement may be found using the formula

$$d = \frac{rh}{H}$$

d is the relief displacement, r is the radial distance of the ground point image from the principal point, h is the elevation of the point from the datum, and H is the flight height above the datum.

$$\begin{aligned}d &= \frac{(3.99 \text{ in})(495 \text{ ft})}{4000 \text{ ft}} \\ &= \frac{1975 \text{ in}}{4000} \\ &= 0.50 \text{ in}\end{aligned}$$

Answer is (D)

OVERLAP AND THE BASE-HEIGHT RATIO

38. What is the typical end-lap and side-lap between stereo pairs?

(A) 60% end-lap
30% side-lap
(B) 50% end-lap
50% side-lap
(C) 40% end-lap
60% side-lap
(D) 10% end-lap
30% side-lap

Solution: The overlapping of adjacent photographs along a flight strip is known as *end-lap* and is usually 60%. The typical 30% overlapping of flight strips is known as *side-lap*.

Answer is (A)

39. Which of the following statements is not a reason for overlapping aerial photographs?

(A) The appearance of the center of each photograph on one or more other photogaphs facilitates the construction of mosaics and minimizes the effects of relief displacement.
(B) Overlapping is necessary for stereoscopic viewing.
(C) Poor photographs may sometimes be eliminated since each area is photographed at least twice.
(D) Overlapping is necessary in order to eliminate the Scheimpflug condition.

Solution: There are several reasons for overlapping aerial photographs, but the Scheimpflug condition is not one of them. The Scheimpflug condition is achieved when the object plane, the image plane, and the plane of the lens intersect along the same line, thus ensuring the sharpness of a projection.

Answer is (D)

40. The base-height ratio is the ratio of the air base, or ground distance between exposure stations, to the flight height above the average terrain elevation. As this ratio increases, which of the following also increases?

(A) the vertical exaggeration in the stereoscopic image from a stereo pair
(B) the sharpness of the focus of the plotting instrument projection
(C) the distance between the isocenter and the principal point
(D) the x-tilt of the photograph

Solution: As the base-height ratio increases, the disparity between the apparent horizontal and vertical scales also increases. The result is that the stereoscopic view of a stereo pair affords an exaggerated depth of the image.

Answer is (A)

41. What is the base-height ratio of an aerial photograph taken with a 6-in focal length lens at 4500 ft above the average elevation of the terrain on a 9-in × 9-in format with a 60% end-lap?

(A) 0.50
(B) 0.60
(C) 0.70
(D) 0.80

Solution: First, calculate the scale of the photograph.

$$S = \frac{f}{H}$$

S is the scale of the photograph, f is the focal length of the lens, and H is the altitude of the aircraft above the average terrain elevation.

$$\begin{aligned} S &= \frac{6 \text{ in}}{4500 \text{ ft}} \\ &= \frac{1 \text{ in}}{750 \text{ ft}} \end{aligned}$$

Next, find the ground length covered by a single photograph.

$$\begin{aligned}\text{ground length} &= (9\text{ in})\left(\frac{750\text{ ft}}{1\text{ in}}\right)\\ &= 6750\text{ ft}\end{aligned}$$

However, with a 60% end-lap, there is a 40% advance between exposures, so the air base is

$$\text{air base} = (0.40)(6750\text{ ft}) = 2700\text{ ft}$$

The base-height ratio is

$$\frac{2700\text{ ft}}{4500\text{ ft}} = 0.60$$

Answer is (B)

GROUND COVERAGE

42. What is the acreage covered by an aerial photograph taken at a height of 12,000 ft above the average ground with a camera having a focal length of 6 in and a 9-in × 9-in format?

(A) 1861 ac
(B) 7438 ac
(C) 4649 ac
(D) 6974 ac

Solution: The first step in solving this problem is to find the scale of the photograph. The formula is

$$S = \frac{f}{H}$$

S is the scale at the valley floor, f is the focal length of the lens, and H is the altitude of the aircraft.

$$\begin{aligned} S &= \frac{6 \text{ in}}{12{,}000 \text{ ft}} \\ &= \frac{1 \text{ in}}{2000 \text{ ft}} \end{aligned}$$

Next, find the ground dimension shown by a 9-in × 9-in photograph.

$$\begin{aligned} \text{ground dimension} &= \left(\frac{2000 \text{ ft}}{1 \text{ in}}\right)(9 \text{ in}) \\ &= 18{,}000 \text{ ft} \end{aligned}$$

Therefore, the required area in acres is

$$\begin{aligned} A &= (18{,}000 \text{ ft})^2 \left(\frac{1 \text{ ac}}{43{,}560 \text{ sq ft}}\right) \\ &= 7438 \text{ ac} \end{aligned}$$

Answer is (B)

43. If aerial photographs have a 9-in format, maintain a 60% end-lap and a 30% side-lap, and are taken at a height of 10,000 ft above the average ground with a camera having a focal length of 8.25 in, what is the acreage covered by the neat model?

(A) 2732 ac
(B) 765 ac
(C) 437 ac
(D) 1338 ac

Solution: The neat model is the portion of a pair of aerial photographs actually used in a mapping project. It is the stereoscopic area between successive principal points, extended perpendicular to the flight direction and ending in the middle of the side-laps.

To find the area of the neat model, begin by finding the scale of the photograph. The formula for the scale is

$$S = \frac{f}{H}$$

S is the scale of the photograph, f is the focal length of the lens, and H is the altitude of the aircraft above the average ground elevation.

$$\begin{aligned} S &= \frac{8.25 \text{ in}}{10{,}000 \text{ ft}} \\ &= \frac{1 \text{ in}}{1212 \text{ ft}} \end{aligned}$$

Next, find the ground dimension of a side of the 9-in × 9-in photograph.

$$\begin{aligned} \text{ground dimension} &= \left(\frac{1212 \text{ ft}}{1 \text{ in}}\right)(9 \text{ in}) \\ &= 10{,}908 \text{ ft} \end{aligned}$$

The region of the overlap used as the neat model is a rectangle 40% of the length and 70% of the length of the photograph. These dimensions can now be translated into ground dimensions.

$$\begin{aligned} \text{ground dimension (base)} &= (0.4)(10{,}908 \text{ ft}) = 4363 \text{ ft} \\ \text{ground dimension (width)} &= (0.7)(10{,}908 \text{ ft}) = 7636 \text{ ft} \end{aligned}$$

The area of the neat model can now be calculated.

$$\begin{aligned} A &= (\text{base})(\text{width}) \\ &= (4363 \text{ ft})(7636 \text{ ft}) \\ &= 33{,}315{,}868 \text{ sq ft} \end{aligned}$$

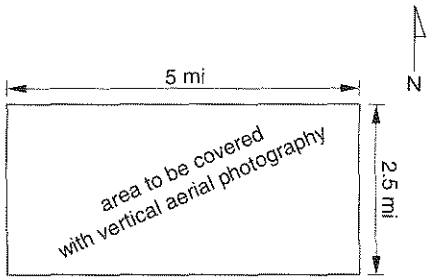
The area can be converted to acreage.

$$\frac{33{,}315{,}868 \text{ sq ft}}{43{,}560 \dfrac{\text{sq ft}}{\text{ac}}} = 765 \text{ ac}$$

Answer is (B)

PLANNING AERIAL PHOTOGRAPHY

The following nine problems refer to the illustration shown.



60% end-lap and 30% side-lap
6-in focal length camera
9-in × 9-in format
scale 1:10,000

Consider the area shown as being virtually level. The coverage of the project with vertical aerial photography should be sufficient to produce a photo-scale of 1:10,000. The end-lap is to be 60% and the side-lap is to be 30% at a minimum. The camera is to have a 6-in focal length and the format is 9 in × 9 in.

44. Given these parameters, what is the ground dimension along the side of a single aerial photograph?

(A) 5250 ft
(B) 3750 ft
(C) 833.33 ft
(D) 7500 ft

Solution: The specified photo-scale is 1:10,000, or

$$\begin{aligned} 1 \text{ in} &= 10{,}000 \text{ in} \\ &= (10{,}000 \text{ in})\left(\frac{1 \text{ ft}}{12 \text{ in}}\right) \\ &= 833.33 \text{ ft} \end{aligned}$$

The format of the photographs is 9 in × 9 in.

$$(9)(833.33 \text{ ft}) = 7500 \text{ ft}$$

Therefore, the ground dimension along the side of a single photograph will be 7500 ft.

Answer is (D)

45. What would be the maximum spacing between adjacent flight lines for the project illustrated?

(A) 7500 ft
(B) 5250 ft
(C) 3750 ft
(D) 2250 ft

Solution: The ground dimension along the side of a single photograph, symbolized as G, is 7500 ft, as shown in Problem 44. Allowing for 30% sidelap, the spacing, symbolized as W, is the remaining 70% of G.

$$\begin{aligned} W &= 0.7G \\ &= (0.7)(7500 \text{ ft}) \\ &= 5250 \text{ ft} \end{aligned}$$

Answer is (B)

46. What arrangement of the flight lines would provide the most efficient coverage of the area illustrated?

(A) The flight lines can be arranged in cardinal directions where convenient, particularly in public lands states where the improvements along section lines provide a guide for the pilot.
(B) It is frequently economical to arrange flight lines parallel with the project boundaries; in this case, the flight lines should be east and west.
(C) A north-south orientation of the flight lines would be most efficient in this instance to accommodate the 60% end-lap.
(D) Both A and B are true.

Solution: In this case, the flight lines would probably be most economically arranged parallel with the longer project boundaries at the north and south.

Answer is (D)

47. Which of the following best accommodates the necessary photo coverage required of the northernmost flight line of the project?

(A) The first, or northernmost, strip should be photographed flying either east or west along the northern boundary of the project.

(B) The first, or northernmost, strip should be photographed flying either east or west along a line about 1500 ft south of the northern boundary of the project.

(C) The first, or northernmost, strip should be photographed flying either east or west along a line about 3750 ft north of the northern boundary of the project.

(D) The first, or northernmost, strip should be photographed flying either east or west along a line about 3750 ft south of the northern boundary of the project.

Solution: The aerial photography should include coverage outside the perimeter of the project, using the minimum number of flight lines. In this case, the first strip can be made by allowing 30% of the photograph to be outside the project boundary. In this instance, G, the ground dimension along one side of the 9-in × 9-in photograph, is 7500 ft.

$$0.3G = (0.3)(7500 \text{ ft}) = 2250 \text{ ft}$$

If 2250 ft of the photograph is outside of the project, then 7500 ft − 2250 ft, or 5250 ft, of the photograph remains inside the project. However, the flight line itself is in the center of the 7500-ft-wide strip, so

$$\begin{aligned} 0.5G - 0.3G &= (0.5)(7500 \text{ ft}) - (0.3)(7500 \text{ ft}) \\ &= 3750 \text{ ft} - 2250 \text{ ft} \\ &= 1500 \text{ ft} \end{aligned}$$

The first, or northernmost, flight line should be 1500 ft south of the northern project boundary.

Answer is (B)

48. If the maximum spacing between east-west flight lines is 5250 ft, and the northern and southern flight lines are each to be 1500 ft inside the project perimeter, what number of flight lines will cover the project, and what will be the adjusted distance between the adjacent flight lines?

(A) four flight lines spaced 4275 ft apart
(B) three flight lines spaced 5100 ft apart
(C) two flight lines spaced 5250 ft apart
(D) four flight lines spaced 3300 ft apart

Solution: The dimension of the project north to south is 2.5 mi, or 13,200 ft. However, the northern and southern flight lines are each 1500 ft inside the perimeter. The distance between the outside flight lines will be

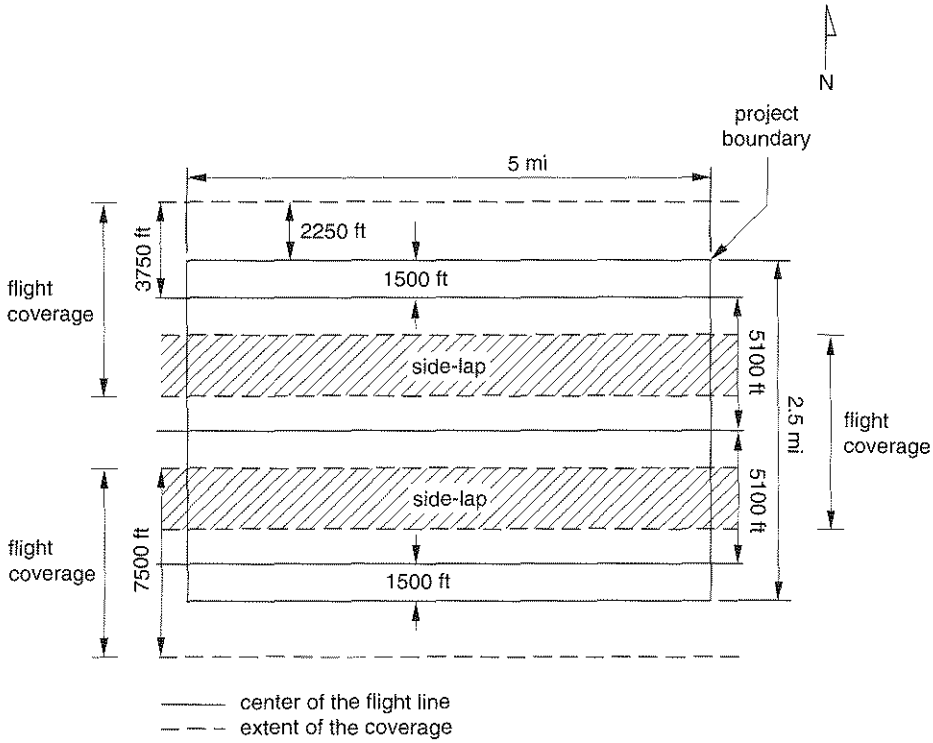
$$\begin{aligned} 13{,}200 \text{ ft} - (2)(1500 \text{ ft}) &= 13{,}200 \text{ ft} - 3000 \text{ ft} \\ &= 10{,}200 \text{ ft} \end{aligned}$$

Since the maximum spacing between flight lines is 5250 ft, one additional flight line is required.

For greatest efficiency, the centers of the flight lines can be adjusted to

$$\frac{10{,}200 \text{ ft}}{2} = 5100 \text{ ft}$$

The total number of flight lines is three, and the two spaces between the centerlines of the adjacent strips are each 5100 ft.



Answer is (B)

49. Allowing for a 60% end-lap, what is the linear advance necessary for each successive photo in the project?

(A) 7500 ft
(B) 5250 ft
(C) 3750 ft
(D) 3000 ft

Solution: The value of G for this project is 7500 ft, which is the ground dimension of a side of the 9-in × 9-in format at the photo-scale of 1:10,000. However, with a 60% end-lap, just 40% of the photo is left. This constitutes the length of the air base.

$$\begin{aligned} B &= \text{linear advance} \\ &= 0.4G \\ &= (0.4)(7500 \text{ ft}) \\ &= 3000 \text{ ft} \end{aligned}$$

Answer is (D)

50. Allowing for four photos outside of the project boundary along each flight line, two on the east and two on the west, how many aerial photographs will be needed to cover the project area?

(A) 52
(B) 47
(C) 39
(D) 21

Solution: The east-west extent of the project is 5 mi, or 26,400 ft. The linear advance for each successive photo along each flight line is 3000 ft, 40% of G, and two photos should be added at the beginning and the end of each flight line to ensure complete coverage. The number of photos per flight line is

$$\frac{26{,}400 \text{ ft}}{3000 \text{ ft}} + 2 + 2 = 8.8 + 2 + 2 = 12.8$$

The closest integral to 12.8 is 13 photographs per strip. There are three flight lines in the project. The total number of photos required is

$$(13)(3) = 39$$

Answer is (C)

51. You are to prepare a guide for the pilot by plotting the flight lines on a quad sheet that has a scale of 1:24,000. What is the spacing between the flight lines on such a plot?

(A) 4.00 in
(B) 5.10 in
(C) 1.67 in
(D) 2.55 in

Solution: The distance between flight lines is 5100 ft. At a scale of 1:24,000, this distance would be

$$\left(\frac{1}{24{,}000}\right)(5100\text{ ft})\left(\frac{12\text{ in}}{\text{ft}}\right) = 2.55\text{ in}$$

Answer is (D)

52. Assuming that the aircraft used in photographing a project travels at a constant speed of 100 mph, how long should the intervalometer setting allow between exposures?

(A) 12 sec
(B) 18 sec
(C) 20 sec
(D) 24 sec

Solution: The rate of the aircraft is more conveniently expressed in feet per second.

$$\left(\frac{100\text{ mi}}{\text{hr}}\right)\left(\frac{5280\text{ ft}}{\text{mi}}\right)\left(\frac{1\text{ hr}}{3600\text{ sec}}\right) = 147\text{ ft/sec}$$

The linear advance (air base) necessary for each successive photo is 3000 ft. The rate of the aircraft is approximately 147 ft/sec, so

$$\frac{3000 \text{ ft}}{147 \dfrac{\text{ft}}{\text{sec}}} = 20.41 \text{ sec}$$

The intervalometer should allow 20 sec between successive photographs.

Answer is (C)

53. Aerial photography for topographic mapping is in the planning stage. The work will be compiled on an optical plotter that can accommodate relief variations in the stereoscopic model of 20% of the projection distance. What is the minimum flight height available in planning the photography of an area that has a maximum difference in elevation of 750 ft between its highest and lowest point?

(A) 3000 ft
(B) 5300 ft
(C) 5500 ft
(D) 3750 ft

Solution: The height of the aircraft above the ground can be treated similarly to the projection distance on the plotter. Where the limitation of the system in handling variations of relief is 20% of the projection distance, the flight height should be at least five times the maximum difference in the elevation of the terrain.

Answer is (D)

PROJECTION PROCEDURES

54. The optical plotting instrument that will be used to compile the aerial photography for a planned project has a C-factor of 800. The project requires a contour interval of 5 ft on the final map. What flight height above the average terrain should be used?

(A) 6000 ft
(B) 5500 ft
(C) 5000 ft
(D) 4000 ft

Solution: The C-factor is an indication of the possible precision available in particular photogrammetric equipment. The larger the C-factor, the higher the available flying heights. The relationship can be expressed as

$$\text{flight height} = (\text{contour interval})(C\text{-factor})$$

In this case, the contour interval is 5 ft and the C-factor is 800.

$$\begin{aligned}\text{flight height} &= (5\text{ ft})(800)\\ &= 4000\text{ ft}\end{aligned}$$

Answer is (D)

55. A wide-angle projection plotter achieves its optimum focus at a projection distance of 30 in when the principal distance is set at 153 mm. What is the focal length of the projector's objective lens?

(A) 216 mm
(B) 8 in
(C) 204 mm
(D) 5 in

Solution: The principal distance is equal to the calibrated focal length of the camera with which the original photograph was taken. In this case, it is 153 mm, or 6 in.

$$(153\text{ mm})\left(0.03937\,\frac{\text{in}}{\text{mm}}\right) = 6.02\text{ in} \quad \text{[round to 6.0 in]}$$

The projection distance is the distance from the lower nodal point of the projector's objective lens to the plane of the optimum focus and, in this case, is 30 in. The projection distances vary with plotters of different design; however, 30 in is typical for wide-angle projectors, and approximately 33 in is typical for normal-angle projectors.

The relationship between the principal and projection distances and the focal length of the projector lens can be expressed as

$$\frac{1}{f} = \frac{1}{p} + \frac{1}{h}$$

f is the focal length of the objective lens of the projector, p is the principal distance, and h is the projection distance.

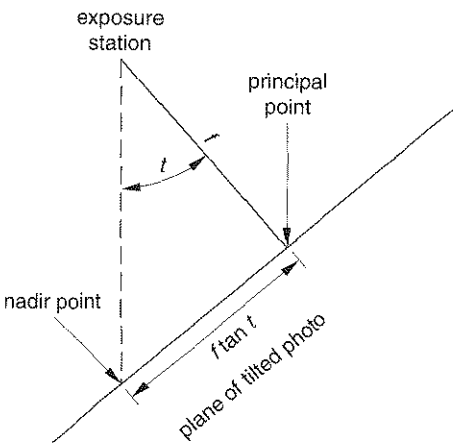
$$\begin{aligned}
\frac{1}{f} &= \frac{1}{6\text{ in}} + \frac{1}{30\text{ in}} \\
&= \frac{5}{30\text{ in}} + \frac{1}{30\text{ in}} = \frac{6}{30\text{ in}} \\
&= \frac{1}{5\text{ in}} \\
f &= 5\text{ in}
\end{aligned}$$

Answer is (D)

56. An aerial photograph having a tilt of 20° was exposed with a 6-in focal length camera. What is the distance between the principal point and the nadir point on the photograph?

(A) 2.18 in
(B) 1.53 in
(C) 0.25 in
(D) 3.62 in

Solution:



The focal length, f, is the distance from the exposure station to the plane of the photograph. If angle t is the tilt, the distance between the nadir point and the principal point is $f \tan t$.

$$(6 \text{ in})(\tan 20°) = 2.18 \text{ in}$$

Answer is (A)

57. In the projection of direct optical stereoplotters, an effort is made to exactly reproduce the relative angular relationship between the two photographs at the moment of exposure. Generally, of the six motions possible with each projector, three are rotations about the mutually perpendicular axes x, y, and z. What is the origin of this system of axes?

(A) the principal point
(B) the focal point
(C) the nadir point
(D) the upper nodal point

Solution: The origin of the axis system in the stereoplotter is the upper nodal point of the projector lens.

Answer is (D)

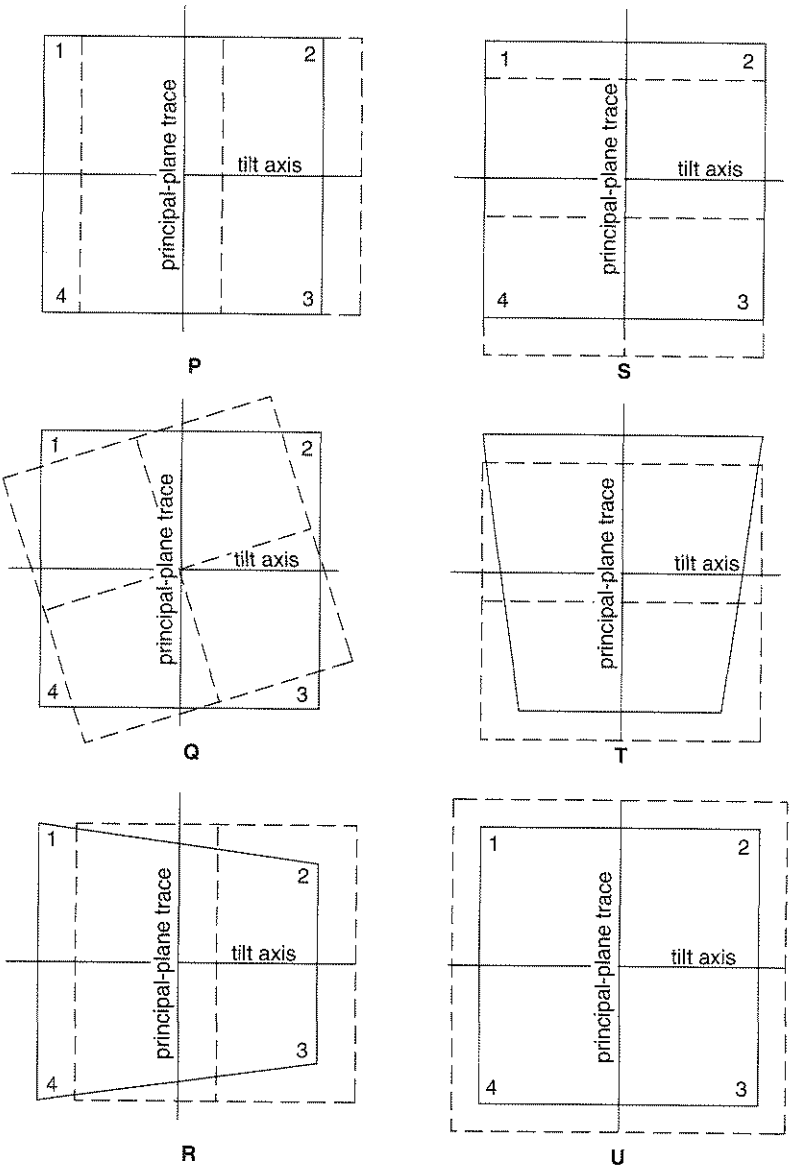
58. Which of the following statements best defines the three linear translations generally available in direct optical stereoplotters?

(A) y-translation changes the spacing between the projectors, z-translation moves the projectors perpendicular to the line between them, and x-translation moves the projectors in relation to the reference table.
(B) x-translation changes the spacing between the projectors, y-translation moves the projectors perpendicular to the line between them, and z-translation moves the projectors in relation to the reference table.
(C) z-translation changes the spacing between the projectors, y-translation moves the projectors perpendicular to the line between them, and x-translation moves the projectors in relation to the reference table.
(D) z-translation changes the spacing between the projectors, x-translation moves the projectors perpendicular to the line between them, and y-translation moves the projectors in relation to the reference table.

Solution: Most direct optical stereoplotters have three angular rotations but not necessarily three linear translations. However, x-translation is essential; it provides the ability to change the spacing between the projectors. z-translation is the adjustment of the distance between the projectors and the reference table.

Answer is (B)

The following six problems refer to the illustrations shown.



A square figure is shown to illustrate the effects of six possible rectifier motions on the projection of a negative or stereomodel, using the one-projector method.

59. Which of the six lettered figures illustrates the effect known as *x-displacement*?

(A) P
(B) S
(C) Q
(D) T

Solution: x-displacement is illustrated in figure P, where the displacement is along the x-axis parallel to the tilt axis.

Answer is (A)

60. Which of the six lettered figures illustrates the effect known as *swing*?

(A) U
(B) Q
(C) T
(D) U

Solution: The rotation around the principal point in figure Q is called *swing*. Another way of describing this motion is rotation about the z-axis.

Answer is (B)

61. Which of the six lettered figures illustrates rotation about the y-axis?

(A) T
(B) Q
(C) R
(D) U

Solution: The apparent convergence of the sides parallel to the tilt axis in figure R is an effect of rotation about the y-axis.

Answer is (C)

62. Which of the six lettered figures illustrates the effect of magnification?

(A) S
(B) Q
(C) P
(D) U

Solution: The radial displacement of all of the corners of the figure relative to the principal point shown in figure U is the effect of magnification.

Answer is (D)

63. Which of the six lettered figures illustrates rotation about the x-axis?

(A) Q
(B) U
(C) P
(D) T

Solution: The apparent convergence of the sides parallel to the principal-plane axis in figure T is an effect of rotation about the x-axis.

Answer is (D)

64. Which of the six lettered figures illustrates the effect known as *y-displacement*?

(A) P
(B) S
(C) Q
(D) T

Solution: y-displacement is illustrated in figure S, where the displacement is along the y-axis parallel to the principal plane.

Answer is (B)

ARTIFICIAL TARGETS

65. Which of the following considerations would be involved in planning the sort of panels, or pre-marks, to use in an aerial photography project?

(A) Does the project include tree-covered areas?

(B) Are there portions of the area to be mapped that are covered with snow or sand?

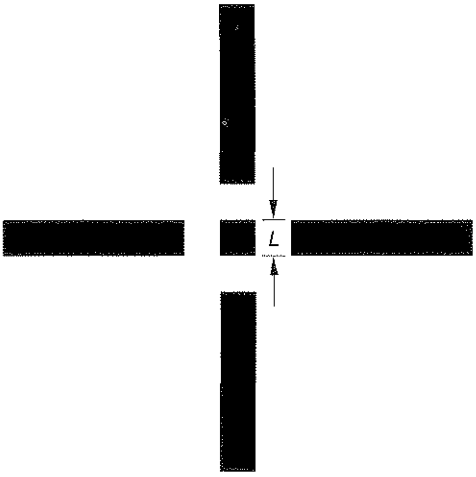
(C) What is the intended photo-scale?

(D) All of the above are appropriate.

Solution: Contrast is one of the primary considerations in choosing the material used in panel points. Light-colored backgrounds require dark pre-marks. Areas of timber or brush raise the possibility that the panels may be obscured by shadows or displacement. The size of the panels is dependent on the intended photo-scale.

Answer is (D)

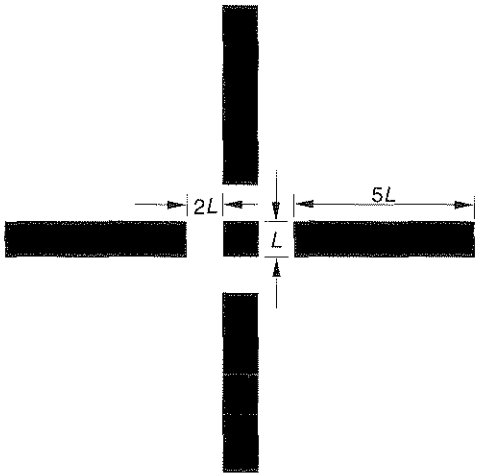
The following two problems refer to the illustration shown.



66. If L is the width of a leg in a typical panel, how long should each leg be?

(A) $5L$
(B) $10L$
(C) $2L$
(D) $20L$

Solution: A length of five times the width of the leg of a panel is generally held to be sufficient.



Answer is (A)

67. The dimensions of the square in the center of the pre-mark panel as shown on the photo negative are required to be 0.02 mm × 0.02 mm. If the focal length of the camera is 6 in and the flight height is 5000 ft above the average terrain elevation, what would be the dimensions of the square on the ground?

(A) 240 mm × 240 mm
(B) $5\frac{3}{4}$ in × $5\frac{3}{4}$ in
(C) 1.3 ft × 1.3 ft
(D) $7\frac{7}{8}$ in × $7\frac{7}{8}$ in

Solution: The scale of the photographs can be expressed as

$$S = \frac{f}{H}$$

S is the scale of the photograph, f is the focal length of the camera, and H is the flight height above the average terrain elevation.

$$S = \frac{0.5 \text{ ft}}{5000 \text{ ft}} = \frac{1}{10{,}000}$$

The scale of the photo will be 1:10,000. If the square must be 0.02 mm × 0.02 mm, then

$$(10{,}000)(0.02 \text{ mm}) = 200 \text{ mm}$$

The ground dimension of the square must be 200 mm × 200 mm.

$$(200 \text{ mm})\left(\frac{1 \text{ in}}{25.4 \text{ mm}}\right) = 7\tfrac{7}{8} \text{ in}$$

Answer is (D)

68. It is common practice for a field crew, using a set of aerial photos, to make positive identification of ground control points. Which of the following procedures is appropriate for such work?

(A) The location of the control point on the photograph should be pricked with a pin.

(B) The photograph should not have holes poked in it; instead, a written description of the position of the control point should be used.

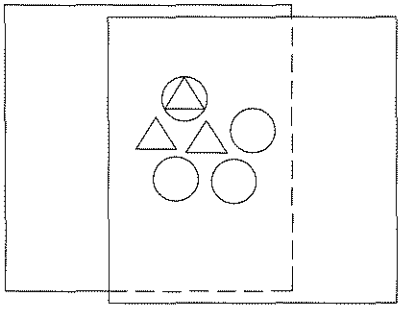
(C) The best identification of a control point is its assigned number and coordinates.

(D) The photographs need not be marked in any way since they are for the reference of the field crew to verify that the control point has not been moved.

Solution: Experience has shown that the most certain method of marking photo control for clear identification is a pin prick penetrating the emulsion at the precise position of the control point. A magnifying glass is useful in this process. Descriptions and sketches of the control are also desirable, but a pin prick is the method least susceptible to misidentification.

Answer is (A)

69. The overlapping aerial photographs shown have the typical symbols used to identify control points in photogrammetry. Which of the following correctly identifies the number and kind of control points illustrated?



(A) 3 horizontal control points
2 vertical control points
1 NGS control point

(B) 3 horizontal control points
2 vertical control points
1 combined horizontal and vertical control point

(C) 3 vertical control points
2 horizontal control points
1 combined horizontal and vertical control point

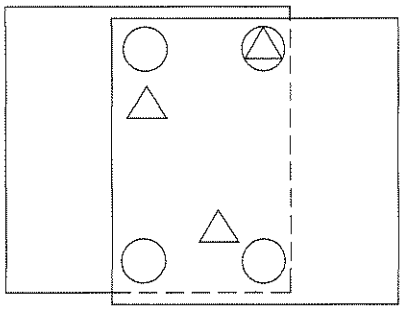
(D) 3 combined horizontal and vertical control points
2 vertical control points
1 horizontal control point

Solution: The usual symbols used to identify control for photogrammetry are the circle, the triangle, and the circle and triangle combined. The circle is used to indicate vertical control. The triangle is used to indicate horizontal control. The combination of the triangle and the circle is used when horizontal and vertical controls are available from the same point.

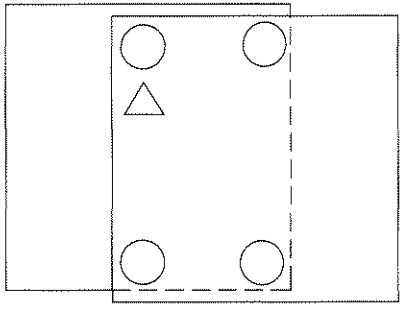
Answer is (C)

70. The following figures illustrate a pair of aerial photographs intended for use in topographic mapping. The photographs show an end-lap of 60%. Which of the figures indicates the best number and configuration of ground control points?

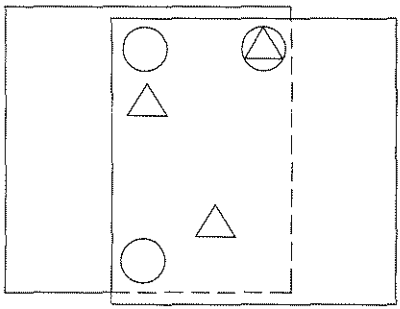
(A)



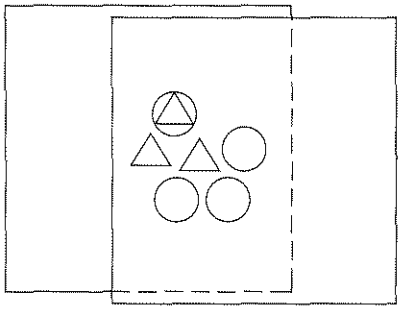
(B)



(C)



(D)



Solution: Topographic mapping requires at least two horizontal and four vertical control points in the overlapping area of adjacent photographs. However, it is best to allow for three horizontal control points rather than the minimum of two. The arrangement of the control is also important. Control in the area that will be included in the side-lap of adjoining flight lines allows for greater efficiency.

Answer is (A)

71. National map standards require that the principal planimetric features on a map of a scale less than 1:20,000 be shown within one-thirtieth of an inch of their true position. Consider a map plotted at a scale of 1 in to 100 ft. What is the maximum allowable error in a ground measurement?

(A) 1.6 ft
(B) 0.11 ft
(C) 3.33 ft
(D) 2.45 ft

Solution: The standard requires the position of features be within one-thirtieth of an inch on the map of their true position on the ground, so

$$\left(\frac{1\text{ in}}{30}\right)\left(\frac{12\text{ in}}{\text{ft}}\right)\left(\frac{100\text{ ft}}{\text{in}}\right) = 40\text{ in} = 3.33\text{ ft}$$

Answer is (C)

72. A widely used rule of mapping is that the photo control should include no error greater than 0.005 of the map scale when that scale is greater than 1:20,000. Considering this rule, what magnitude of error in ground control may be tolerated for a map of the scale 1:24,000?

(A) 1 ft
(B) 15 ft
(C) 7.23 ft
(D) 10 ft

Solution: The scale 1:24,000 may also be stated as 1 in = 2000 ft.

$$(2000\text{ ft})(0.005) = 10\text{ ft}$$

Answer is (D)

73. Which of the following specifications is good practice in preparing a pre-mark panel to function as a vertical control point for photogrammetry?

(A) The panel should be set in a flat area, and the elevation of the monument at its center determined.
(B) A description and sketch of the panel should be prepared.
(C) The vertical measurement from the monument to the panel itself should be made if the latter is above or below the monument.
(D) All of the above are true.

Solution: Finding the difference in elevation between the monument set at the center of a pre-mark panel point and the panel material itself is sometimes neglected. The panel is the basis of the photogrammetrist's determination of elevation. If its elevation differs from that of the monument, the difference should be known.

Answer is (D)

74. National map standards require that the elevations of 90% of the points tested be correct within what part of its contour interval?

(A) one-sixth
(B) one-fifth
(C) one-fourth
(D) one-half

Solution: The national map standards call for vertical accuracy to be within one-half of the contour interval shown on the map. A more typical standard for vertical control in photogrammetry is one-fifth of the contour interval.

Answer is (D)

75. Which of the following best describes the meaning of the term *picture point* as it is used in photogrammetry?

(A) a particularly clear pre-marked artificial panel point on an aerial photograph

(B) a pin prick on the emulsion of an aerial photograph made during control verification

(C) a sharp and unambiguous natural feature used to supplement ground control

(D) any one of a number of points imagined to exist on aerial photographs, including the nadir point, the principal point, and the isocenter, among others

Solution: Picture points are generally natural features that are well defined on aerial photographs under magnification. They must be accessible and capable of positive identification in the field as well. When such natural features fall in an appropriate position on the aerial photographs, picture points are frequently used to augment artificially set pre-marked photogrammetric control.

Answer is (C)